

- Web-based materials/chemistry simulation platform

Materials Square Class Curriculum

○



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[Lab.3] Oxygen Reduction Reaction (ORR)

Learning Objectives:

→ Verify the Oxygen Reduction Reaction (ORR) process on the electrocatalysts using density functional theory (DFT) calculations.

Materials: Platinum (111) surfaces



MatSQ Credit: ~ 172 CPU hour (\$ 43.0)

[Lab.3] Oxygen Reduction Reaction (ORR)

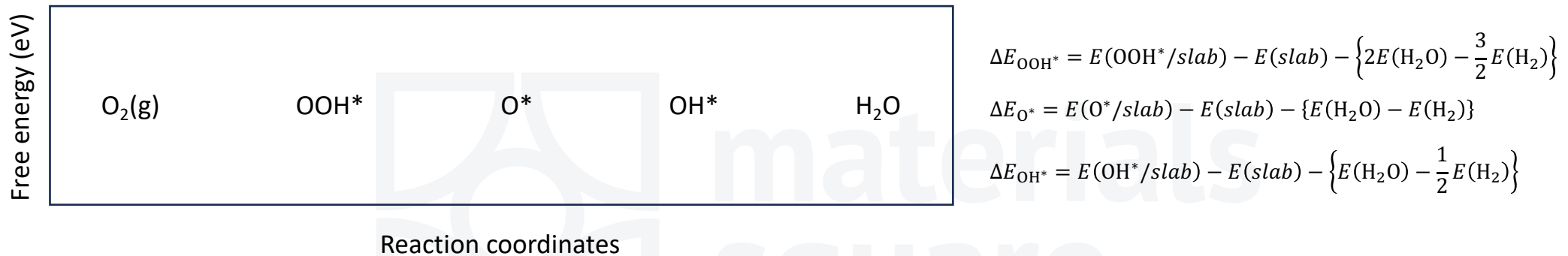
Pt(111) surfaces

1. Build a Pt(111) surface structure
2. Optimize the atomic structure of Pt(111) surface
3. Build adsorption structures
4. Optimize atomic structures (adsorption structures)
5. Analyze optimized adsorption structures
6. Build molecule structures
7. Optimize molecule structures using High Throughput
8. Calculate adsorption energies
9. Calculate free-energy diagram

MatSQ Credit: ~ 172 CPU hour (\$ 43.0)

[Lab.3] Oxygen Reduction Reaction (ORR)

ORR Free energy diagram



List of required calculations

- Molecules : H_2 , H_2O ,
- Substrate: Pt(111)
- Adsorption structures: $O_2/Pt(111)$, $OOH/Pt(111)$, $O/Pt(111)$, $OH/Pt(111)$

[Lab.3] Oxygen Reduction Reaction (ORR)

1. Build a Pt(111) surface structure

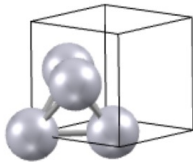
Structure Builder

Modeling Cell Atom Extension

STRAIN CLONE VACUUM CLEAVE MERGE PRIMITIVE CONVENTIONAL

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal



Cleave Surface

(h,k,l) = 1 1 1

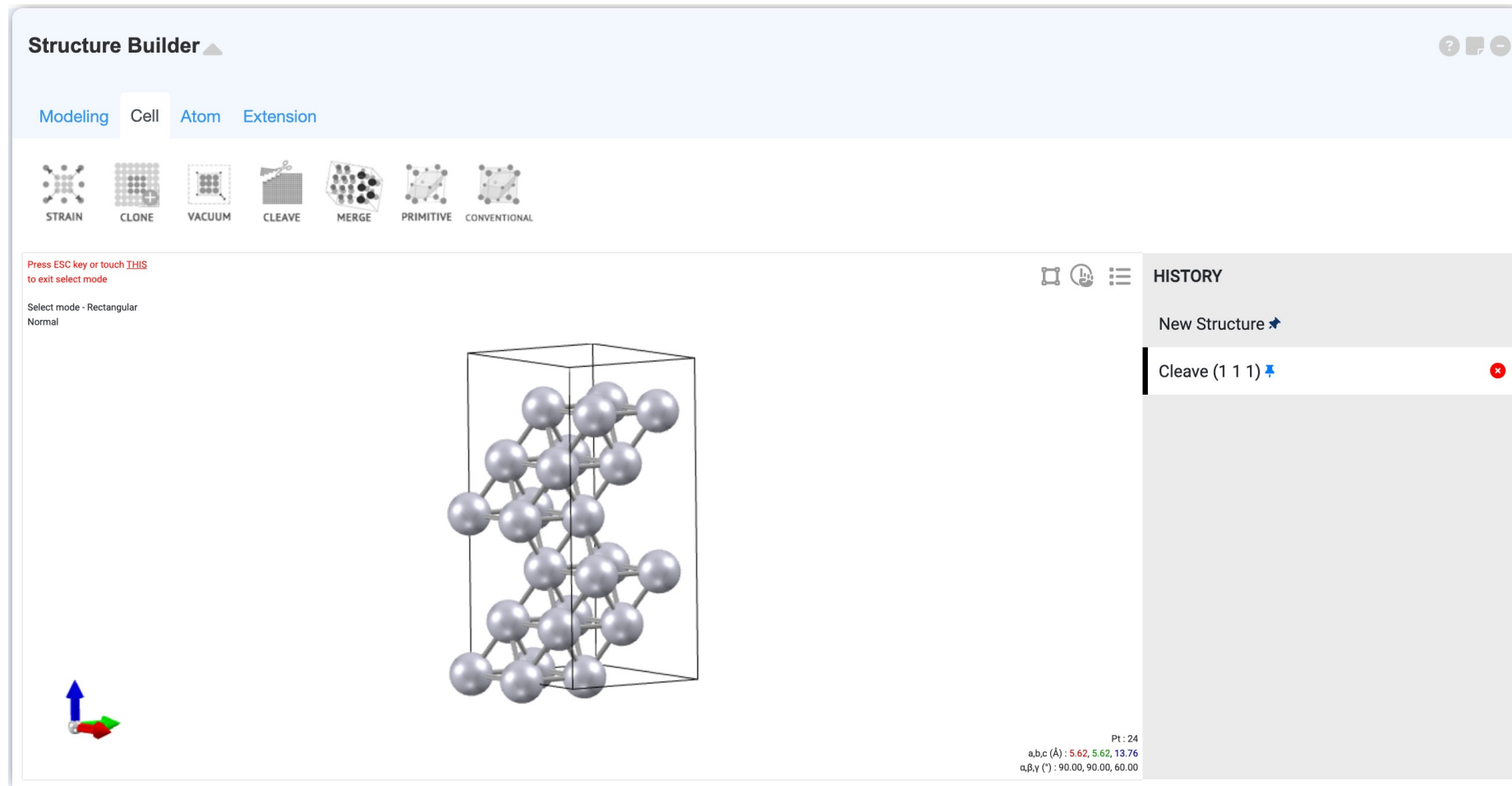
(max c length Å) 30 ✓ Apply

no.	a	b	c	α	β	γ
1	5.619	5.619	5.619	60.000	60.000	60.000
2	5.619	5.619	9.733	73.221	90.000	60.000
3	5.619	5.619	13.764	90.000	90.000	60.000
4	5.619	5.619	18.636	90.000	98.671	60.000
5	5.619	5.619	23.168	83.035	90.000	60.000
6	5.619	5.619	28.095	84.261	95.739	60.000

a,b,c (Å) : 3.97, 3.97, 3.97
 α,β,γ (°) : 90.00, 90.00, 90.00

[Lab.3] Oxygen Reduction Reaction (ORR)

1. Build a Pt(111) surface structure



[Lab.3] Oxygen Reduction Reaction (ORR)

1. Build a Pt(111) surface structure

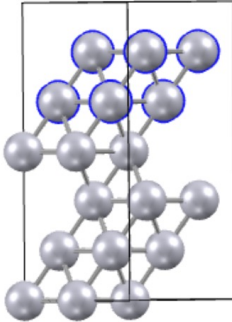
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular
Normal
Pt=8



HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt: 24
a,b,c (Å): 5.62, 5.62, 13.76
α,β,γ (°): 90.00, 90.00, 60.00

[Lab.3] Oxygen Reduction Reaction (ORR)

1. Build a Pt(111) surface structure

Structure Builder

Modeling **Cell** Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch THIS to exit select mode

Select mode - Rectangular Normal

Unit: Angstrom(Å)

Cell: Lattice Parameters

a	5.6190615219	α (°)	90
b	5.6190615219	β (°)	90
c	20	γ (°)	59.999999999

Atoms: Cartesian ☐ Selected Atoms

ID	EL.	x	y	z
1	Pt	0	0	0
2	Pt	5.6190	3.2441	2.2939
3	Pt	2.8095	1.6220	4.5879
4	Pt	0	0	6.8819
5	Pt	2.8095	0	0

+ Add Atom

[Lab.3] Oxygen Reduction Reaction (ORR)

2. Optimize the atomic structure of Pt(111) surface

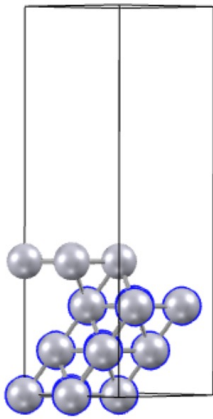
Structure Builder

Modeling Cell Atom Extension

DATABASE JOBS MODULE FILE CRYSTAL SLAB PRESET DRAW MOLECULE EDIT

Press ESC key or touch **THIS** to exit select mode

Select mode - Rectangular
Normal
Pt=12



HISTORY

- New Structure
- Cleave (1 1 1)
- Manual Edit

Pt : 16
a,b,c (Å) : 5.62, 5.62, 20.00
α,β,γ (°) : 90.00, 90.00, 60.00

[Lab.3] Oxygen Reduction Reaction (ORR)

2. Optimize the atomic structure of Pt(111) surface

Quantum Espresso

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info

Target property
Energy / Structure

&CONTROL

Calculation type
scf

input.pw.x

Potential

&CONTROL
calculation = 'scf'
outdir = './output/'

	Checkbox	Input	Selected
Fix			
No.	Ele.	☐	☐ x ☐ y ☐ z
1	Pt	☒	☒ 0.000 ☒ 0.000 ☒ 0.000
2	Pt	☒	☒ 5.619 ☒ 3.244 ☒ 2.294
3	Pt	☒	☒ 2.810 ☒ 1.622 ☒ 4.588
4	Pt	☐	☐ 0.000 ☐ 0.000 ☐ 6.882
5	Pt	☒	☒ 2.810 ☒ 0.000 ☒ 0.000
6	Pt	☒	☒ 2.810 ☒ 3.244 ☒ 2.294
7	Pt	☒	☒ 5.619 ☒ 1.622 ☒ 4.588
8	Pt	☐	☐ 2.810 ☐ 0.000 ☐ 6.882
9	Pt	☒	☒ 1.405 ☒ 2.422 ☒ 0.000

Structure optimization
No optimization ▼

&ELECTRONS
Max iteration step 100
SCF must converge True ▼
Mixing beta 0.5
KPOINTS
Sampling Monkhorst-Pack ▼
Grid 3 3 1
Shift 0 0 0

Job Submit
Resource : Spot ▼
PPN : 48 ▼ ram : 192 GB

Keyword information

[Lab.3] Oxygen Reduction Reaction (ORR)

2. Optimize the atomic structure of Pt(111) surface

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info Update Structure

Keyword information

Target property

Precision

Model type

Structure optimization

Option

Energy / Structure

High (long time)

Isolate System

Ionic optimization

☐ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

relax

200

high

0.00038

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

40

320

smearing

0.002

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

KPOINTS

Sampling

Grid

Shift

200

True

plain

0.5

0.000001

random

Monkhorst-Pack

551

000

&IONS

Ion dynamics

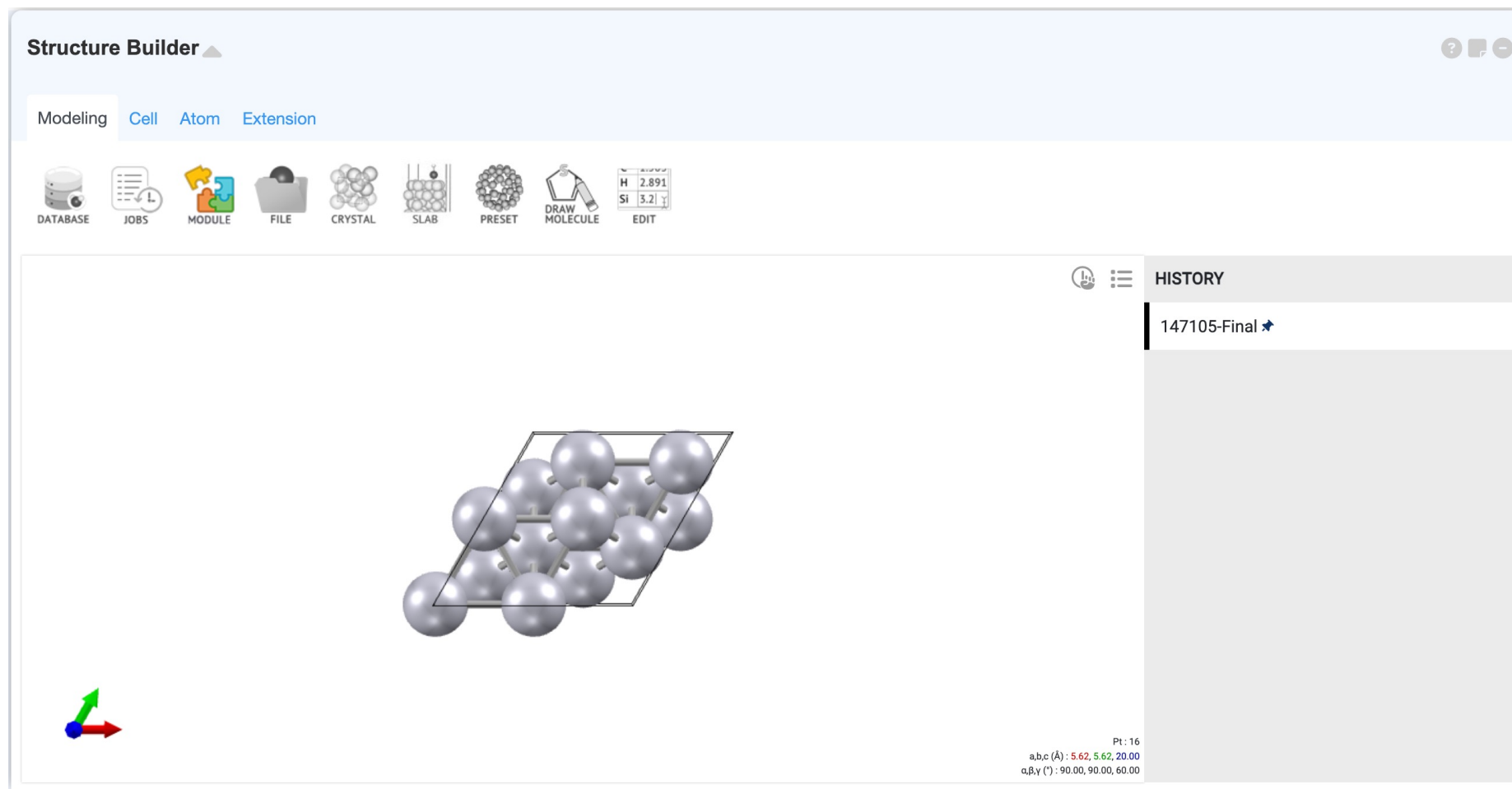
Upscale

bfgs

100

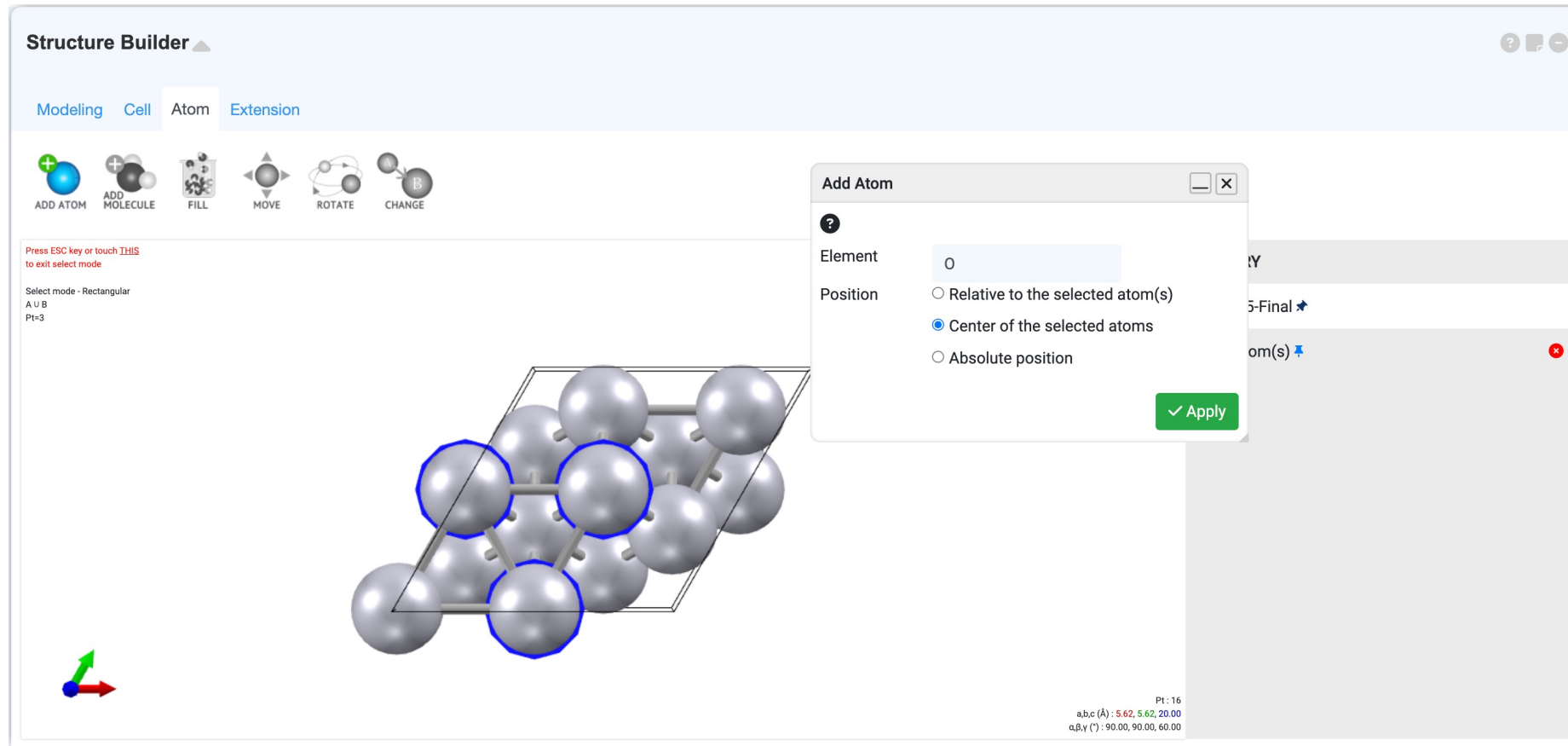
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



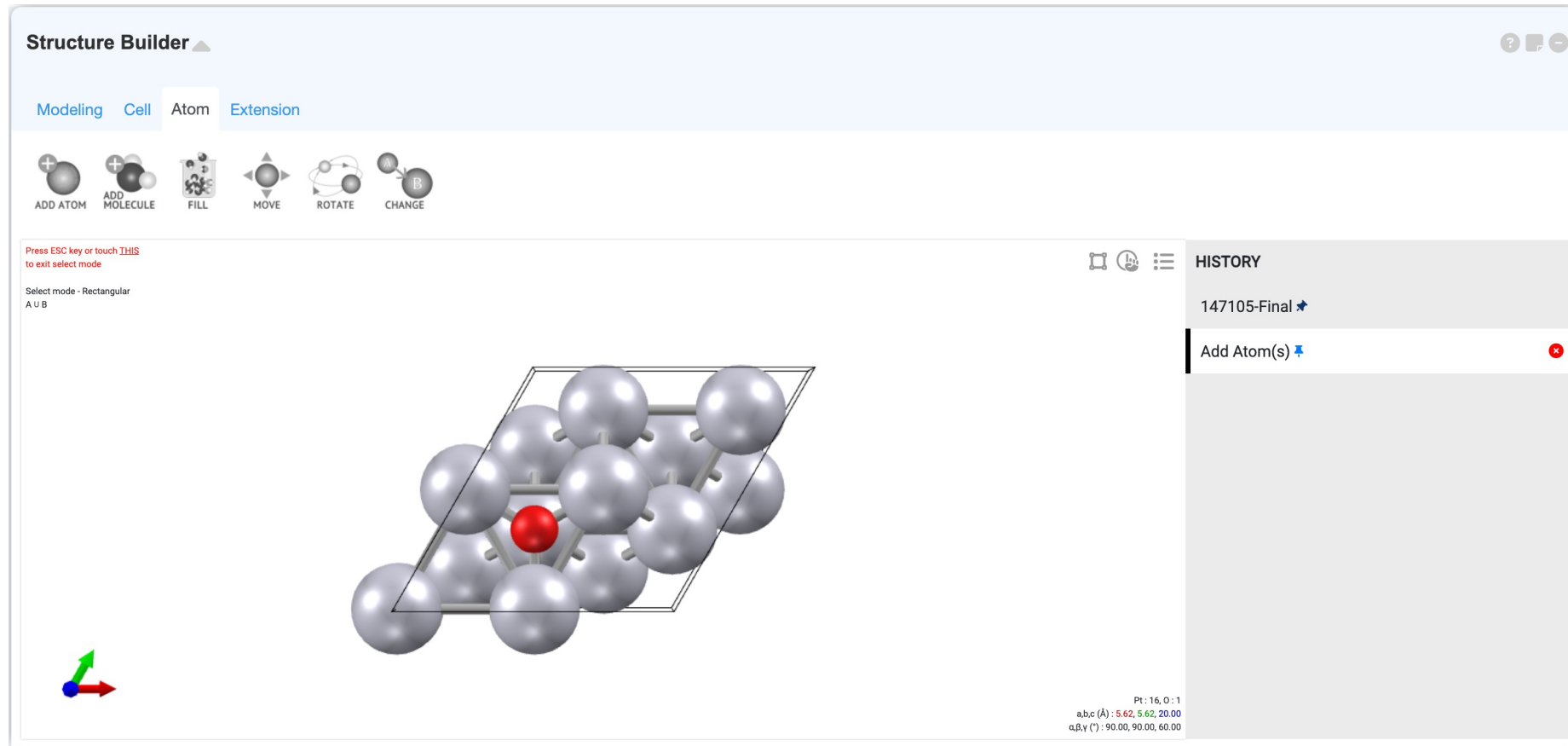
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



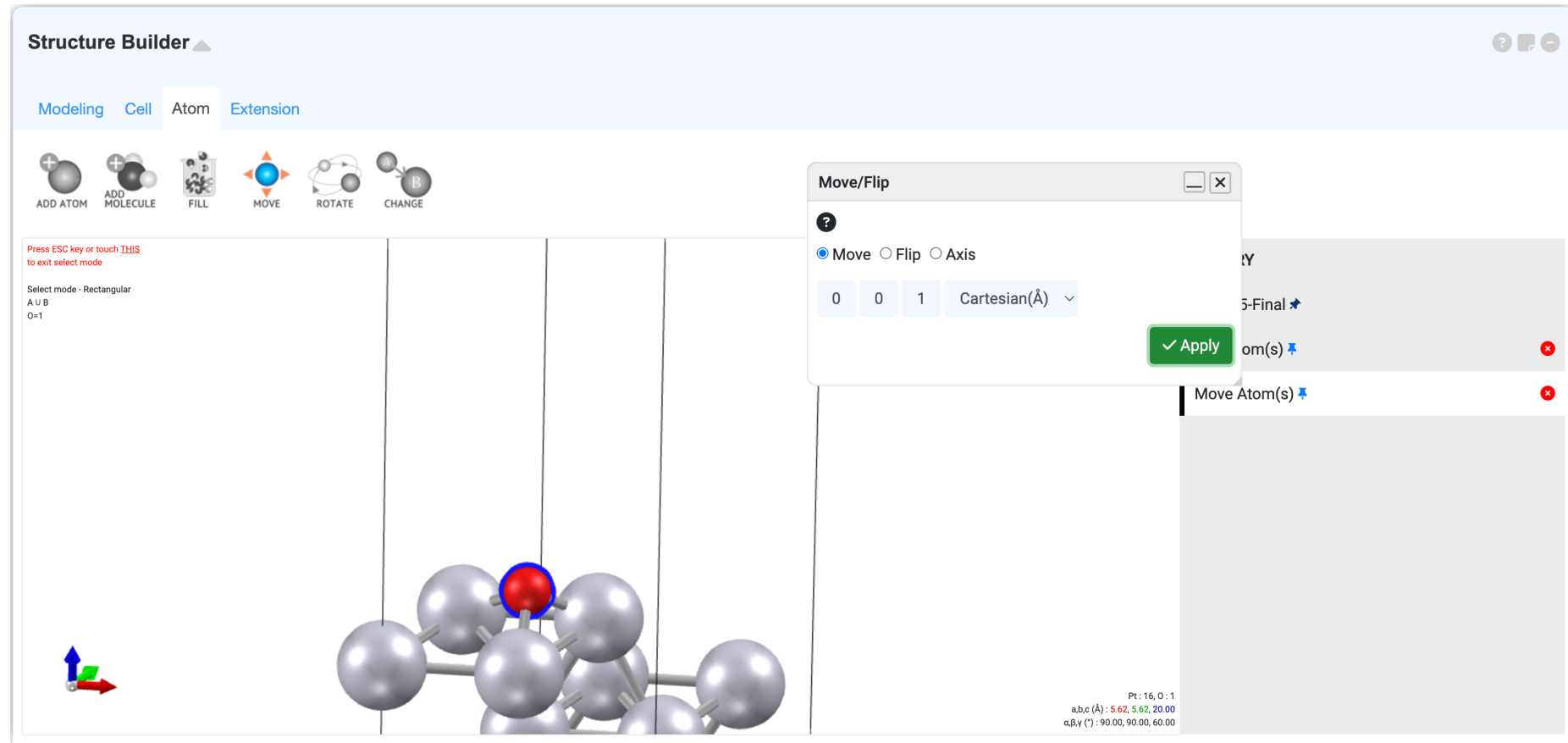
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



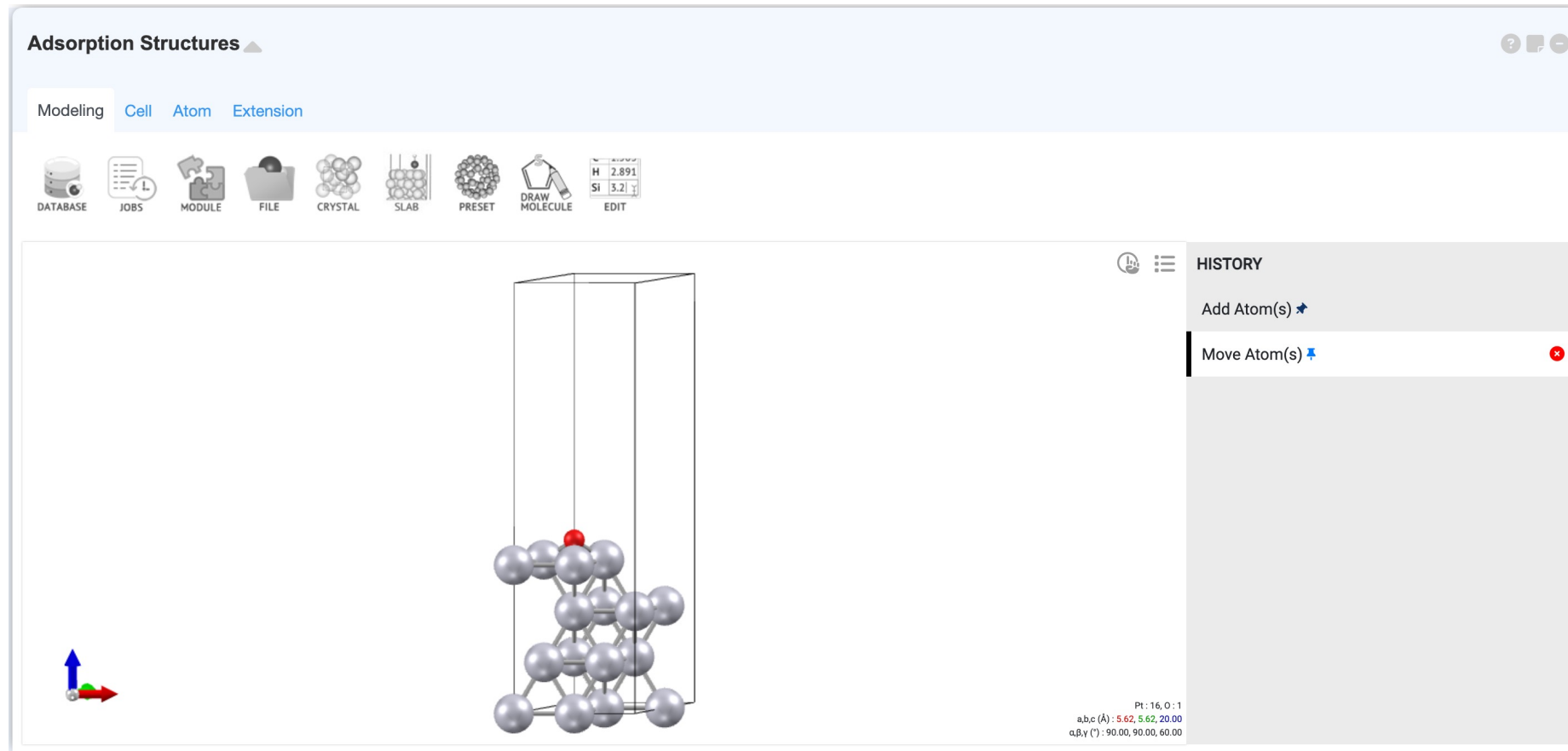
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



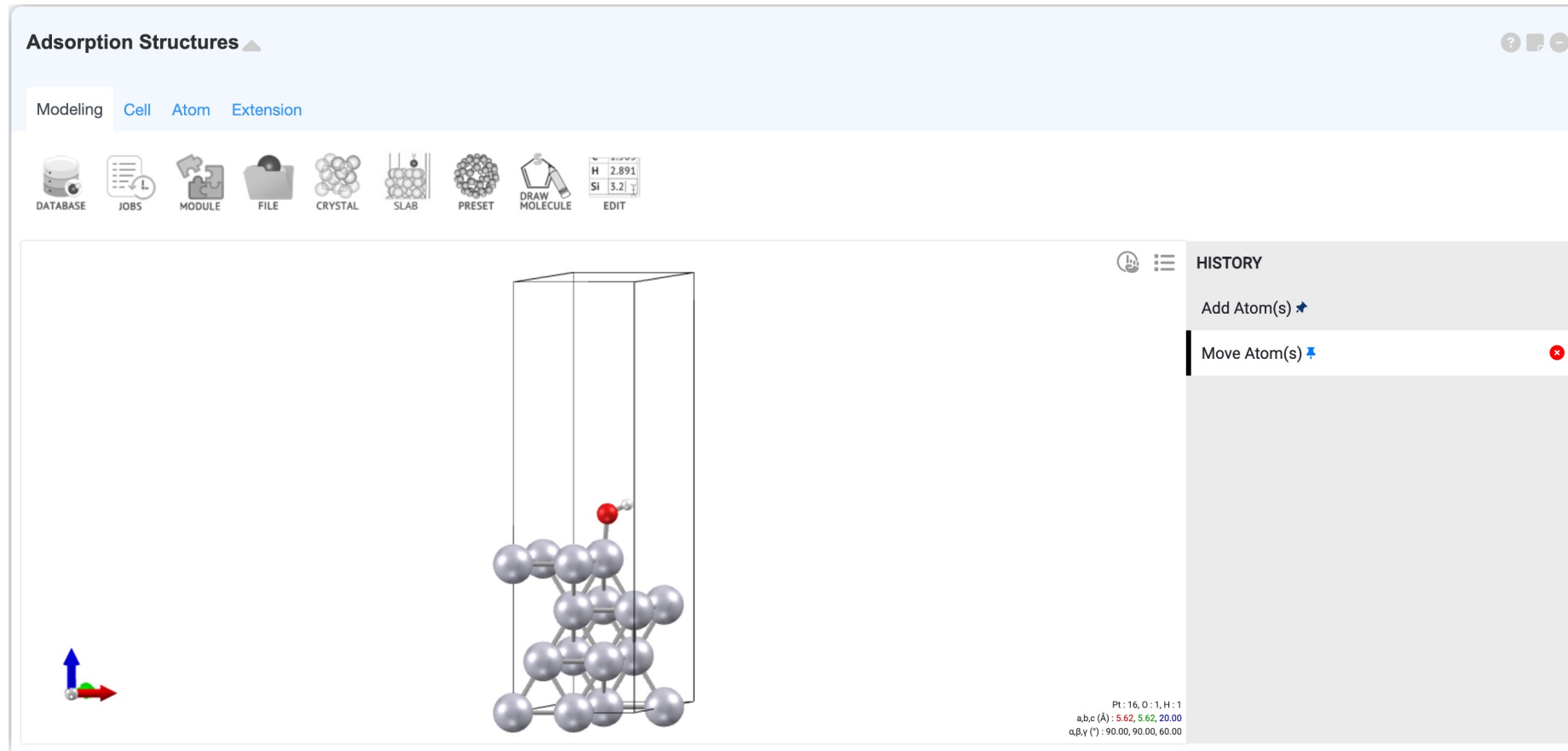
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



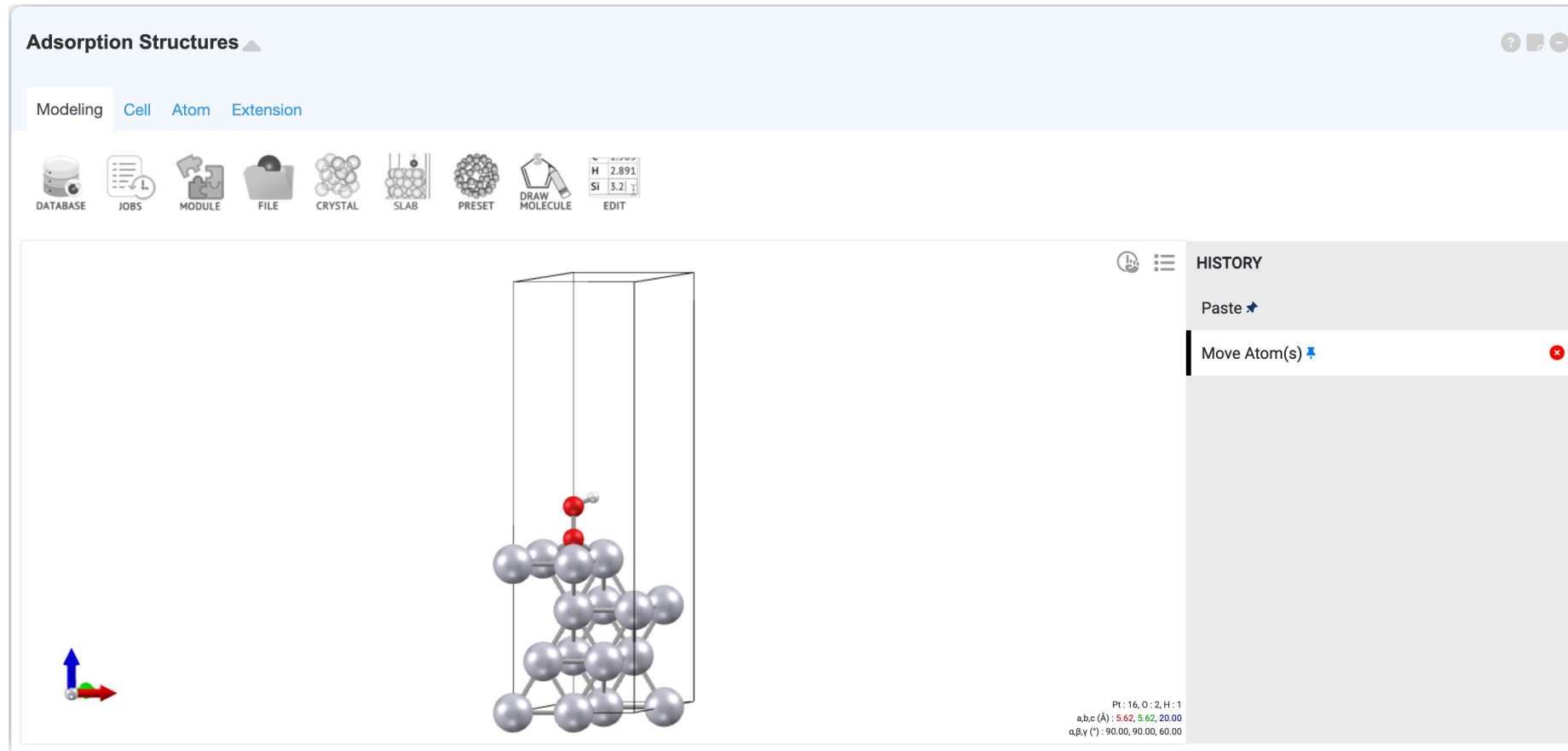
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



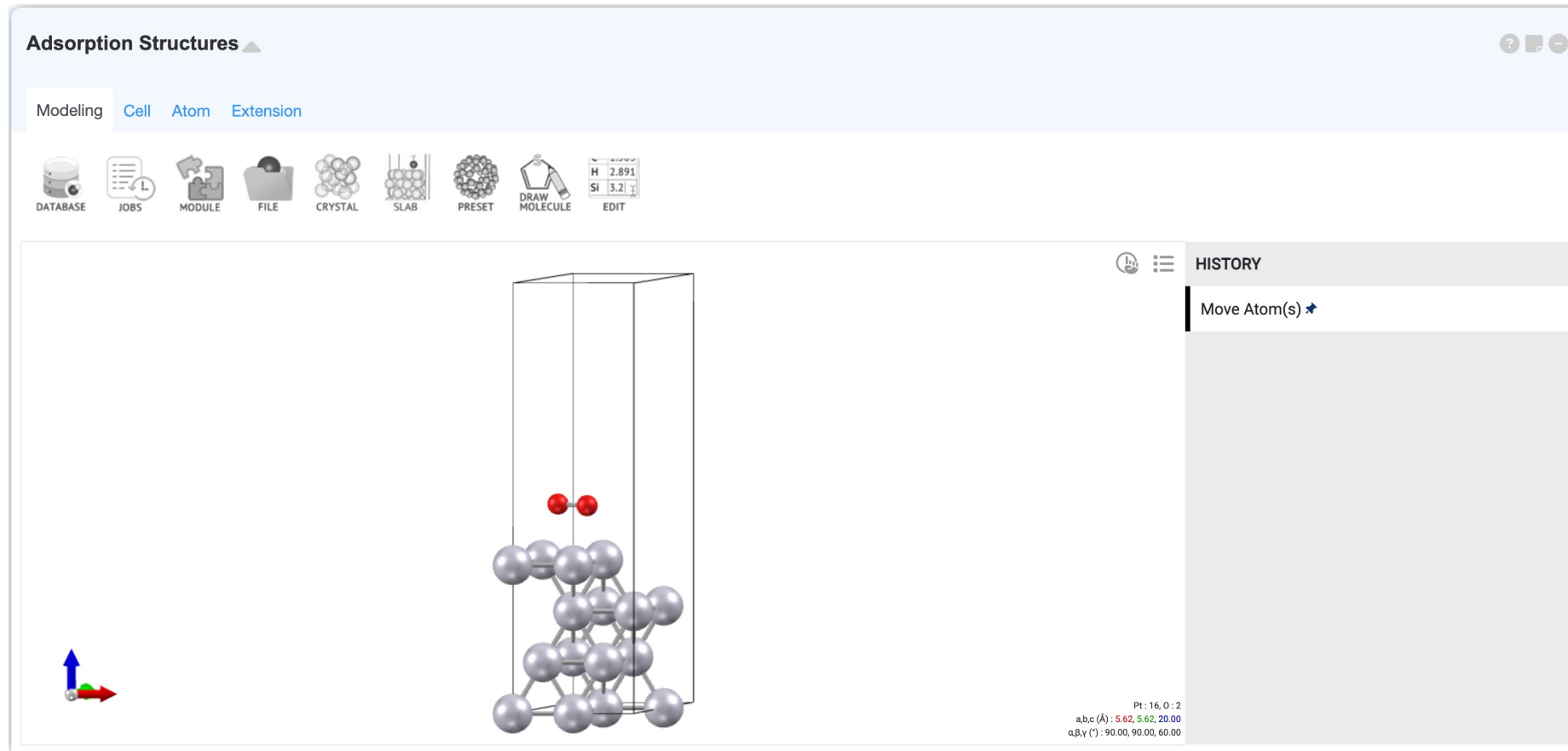
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



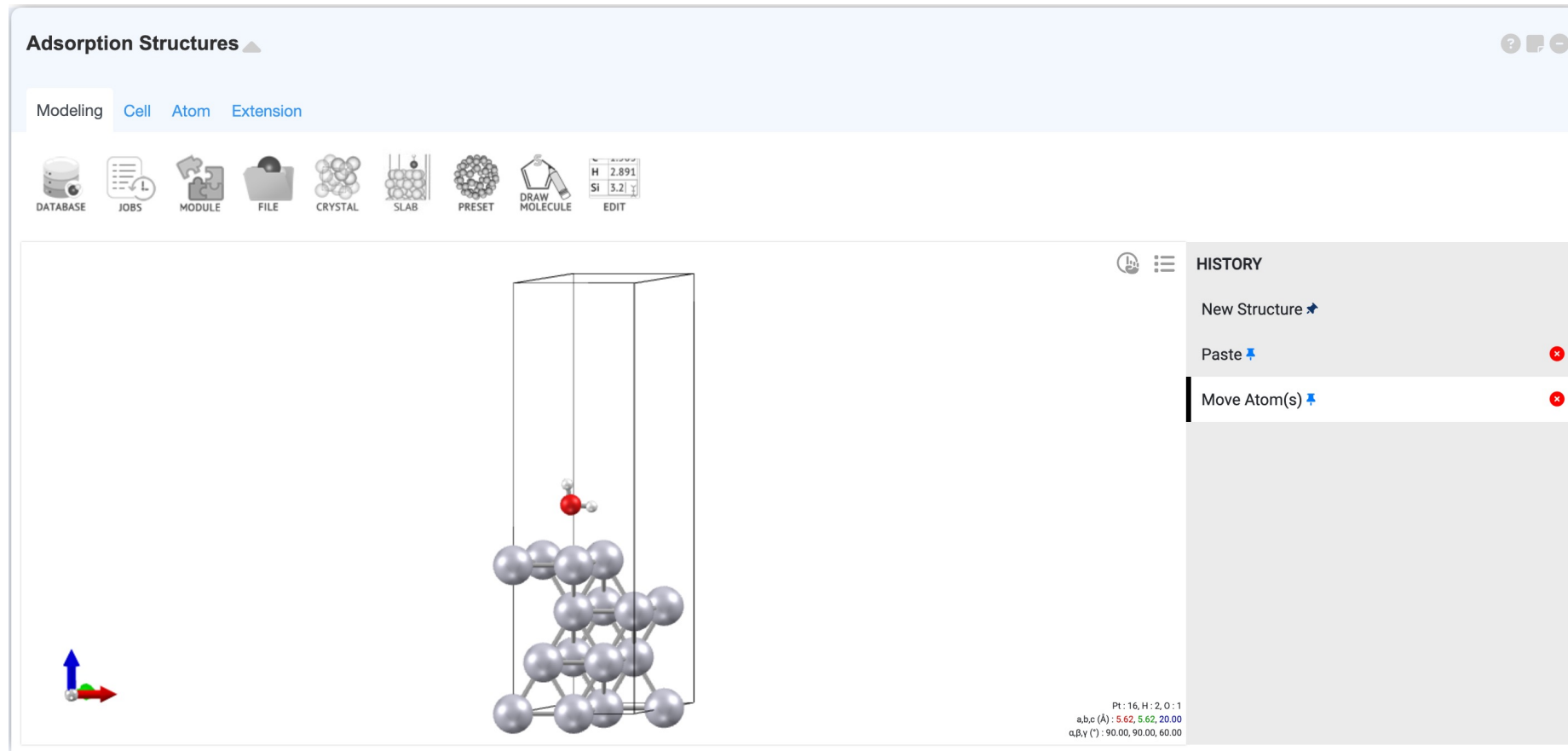
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



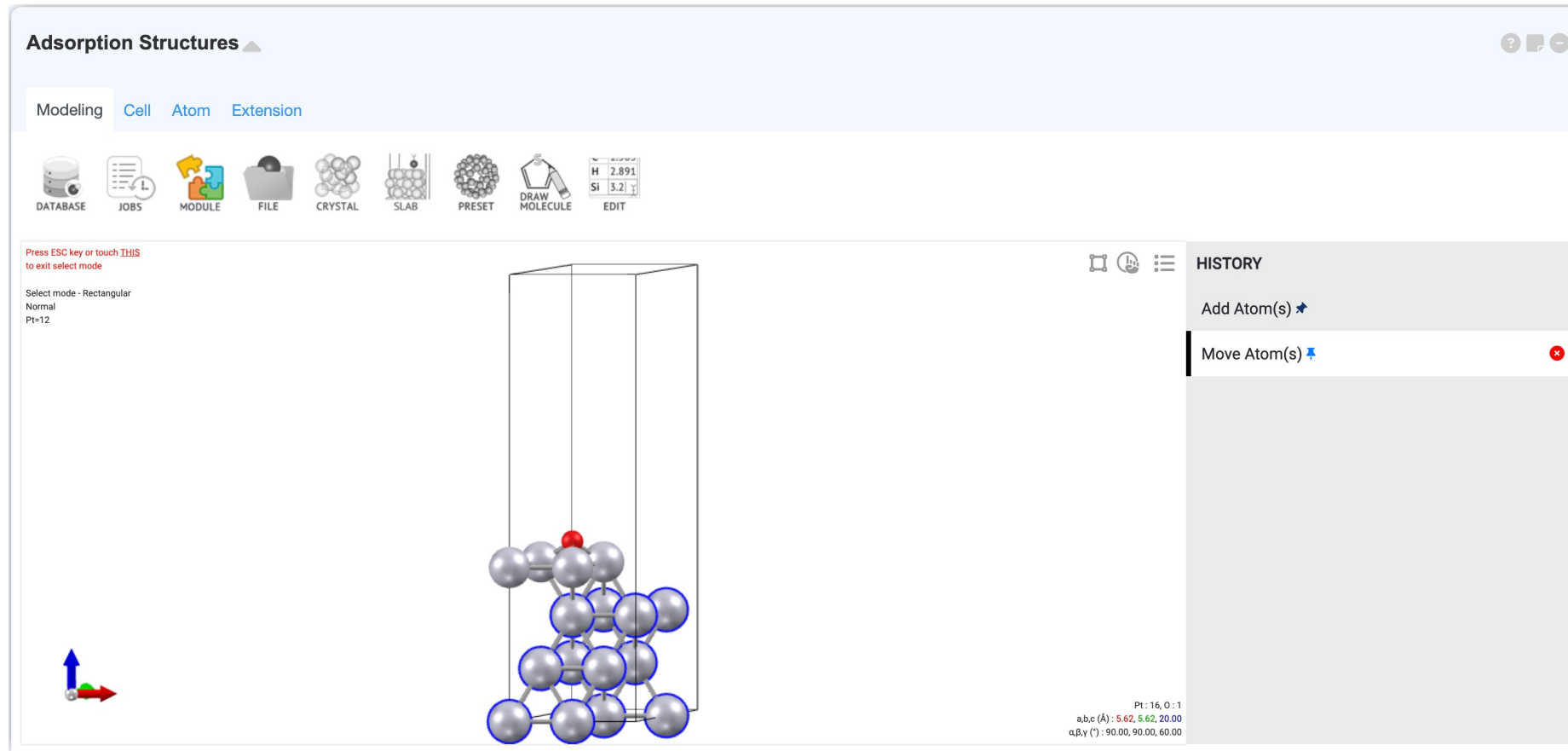
[Lab.3] Oxygen Reduction Reaction (ORR)

3. Build adsorption structures



[Lab.3] Oxygen Reduction Reaction (ORR)

4. Optimize atomic structures (adsorption structures)



[Lab.3] Oxygen Reduction Reaction (ORR)

4. Optimize atomic structures (adsorption structures)

Quantum Espresso

PWscf + DOS

Scripting Option : Template Atom info

Target property
Energy / Structure

&CONTROL
Calculation type

input.pw.x Potential

&CONTROL
calculation = 'scf'
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'from_scratch'

	Checkbox	Input	Selected
		Fix	
	No.	Ele.	☐ x ☐ y ☐ z
1	Pt	☒ 0.000	☒ 0.000 ☒ 0.000
2	Pt	☒ 5.619	☒ 3.244 ☒ 2.294
3	Pt	☒ 2.810	☒ 1.622 ☒ 4.588
4	Pt	☐ 0.000	☐ 0.000 ☐ 6.905
5	Pt	☒ 2.810	☒ 0.000 ☒ 0.000
6	Pt	☒ 2.810	☒ 3.244 ☒ 2.294
7	Pt	☒ 5.619	☒ 1.622 ☒ 4.588
8	Pt	☐ 2.810	☐ 0.000 ☐ 6.905
9	Pt	☒ 7.024	☒ 2.422 ☒ 0.000

30
smearing
0.002

Structure optimization
No optimization

&ELECTRONS
Max iteration step
SCF must converge
Mixing beta
KPOINTS
Sampling
Grid
Shift

100
True
0.5
Monkhorst-Pack
3 3 1
0 0 0

Job Submit
Resource : Spot
PPN : 48 ram : 192 GB
Node type : Computing

[Lab.3] Oxygen Reduction Reaction (ORR)

4. Optimize atomic structures (adsorption structures)

Quantum Espresso ▲

PWscf + DOS ▼

Scripting Option : Template ▼ Atom info Update Structure

Target property

Precision

Model type ?

Structure optimization

Option

Energy / Structure ▼

High (long time) ▼

Isolate System ▼

Ionic optimization ▼

☒ Spin polarization

☐ Van der Waals

☐ Electric field

☐ Spin Orbit Coupling

☐ DFT+U

&CONTROL

Calculation type

Max SCF steps

Information amount

Force threshold (Ry/bohr)

relax ▼

200

high ▼

0.001

&SYSTEM

E_{cut}(wfc)(Ry)

E_{cut}(rho)(Ry)

Occupations

Gaussian broadening (Ry)

Number of electron spin

Starting magnetization

Pt

O

40

320

smearing ▼

0.002

2 (up,down) ▼

1

1

&ELECTRONS

Max iteration step

SCF must converge

Mixing mode

Mixing beta

Convergence threshold (Ry)

Starting wavefunction

200

True ▼

plain ▼

0.5

0.000001

random ▼

&IONS

Ion dynamics

Upscale

bfgs ▼

100

KPOINTS

Sampling

Grid

Shift

Monkhorst-Pack ▼

5

5

1

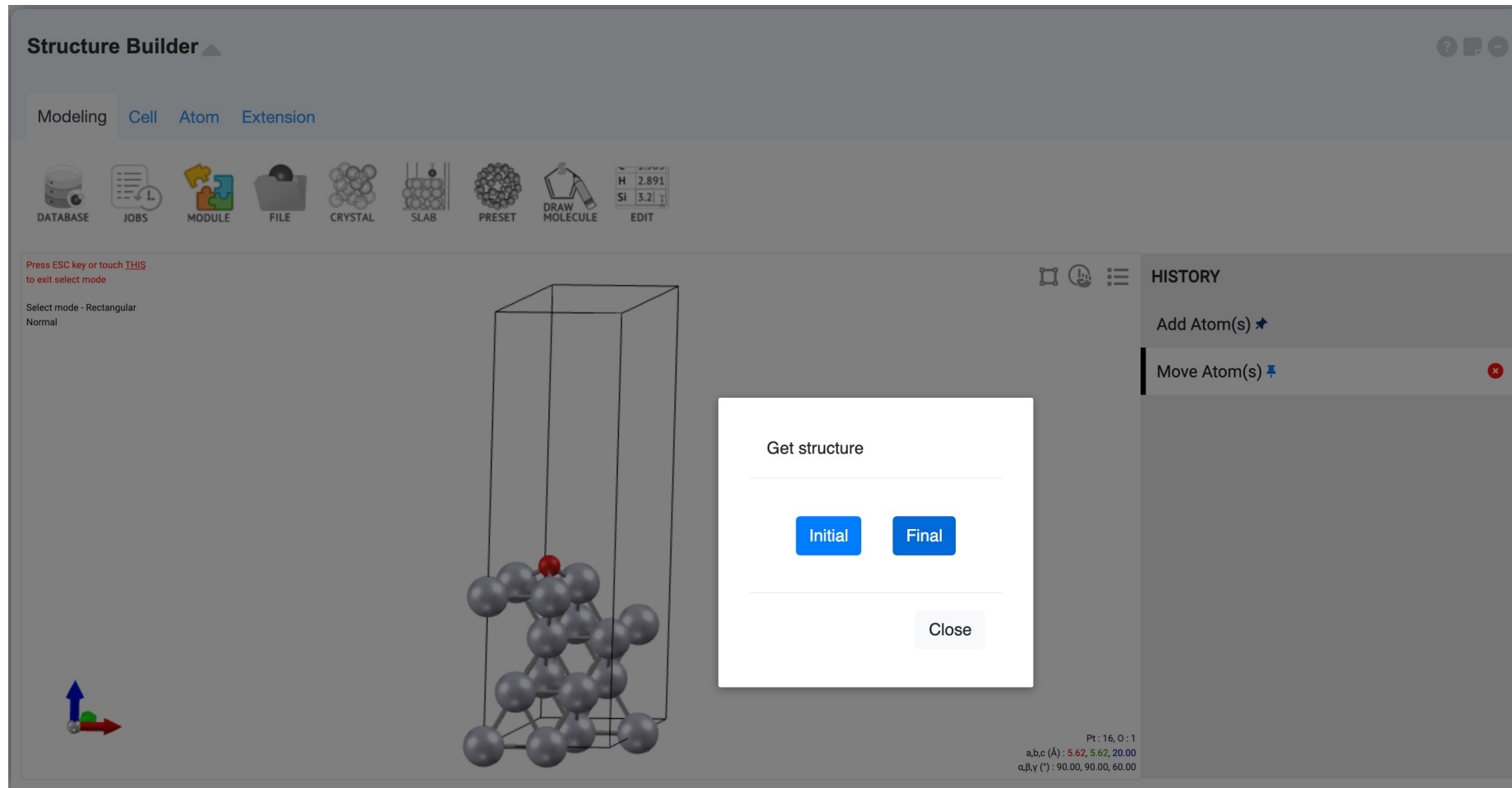
0

0

0

[Lab.3] Oxygen Reduction Reaction (ORR)

5. Analyze optimized adsorption structures



Energy ▲

Energy (eV)

Job	Color	Final Energy
14710	Red	-162675.00660 eV
147106	Orange	-163240.60209 eV
147109	Yellow	-163257.36957 eV

[Lab.3] Oxygen Reduction Reaction (ORR)

6. Build molecule structures

Molecules  

Modeling **Cell** Atom Extension

 DATABASE  JOBS  MODULE  FILE  CRYSTAL  SLAB  PRESET  DRAW MOLECULE  EDIT





Manual Edit 

Unit: Angstrom(Å) 

Cell  Lattice Vectors 

a	15	0	0
b	0	15	0
c	0	0	15

Atoms  Cartesian  ☐ Selected Atoms

ID	EL.	x	y	z	
1	H	6.75	7.5	6.75	
2	H	7.95	7.5	6.75	

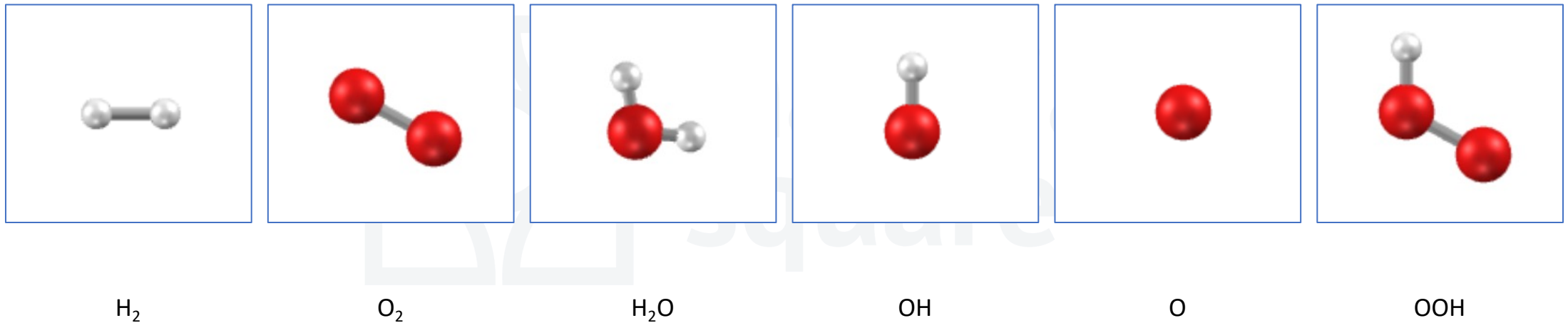
 Add Atom

a,b,c (Å): 15.00,
α,β,γ (°): 90.00,

 Reset  Apply

[Lab.3] Oxygen Reduction Reaction (ORR)

6. Build molecule structures



[Lab.3] Oxygen Reduction Reaction (ORR)

7. Optimize molecule structures using High Throughput (molecules)

High Throughput

Default

PWscf

```
&CONTROL
calculation = 'relax'
forc_conv_thr = 0.00038
nstep = 200
outdir = './output/'
prefix = 'VLAB'
pseudo_dir = './'
restart_mode = 'from_scratch'
tpnfor = .TRUE.
verbosity = 'high'
wf_collect = .TRUE.
/
```

Update

Element	Filename
H	H.pbe-kjpaw_psl.1.0.0.UPF
O	O.pbe-n-kjpaw_psl.0.1.1.UPF

Calculations

Waiting x 6

Name	Formula	Script
H2	H2	pw.x
O2	O2	pw.x
H2O	H2O1	pw.x
O	O1	pw.x
OH	O1H1	pw.x
OOH	O2H1	pw.x

StructureScriptPotential

[Lab.3] Oxygen Reduction Reaction (ORR)

8. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{surfaces}})]$$



[Lab.3] Oxygen Reduction Reaction (ORR)

8. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{surfaces}})]$$



[Lab.3] Oxygen Reduction Reaction (ORR)

8. Calculate adsorption energies

$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{surfaces}})]$$

Memo

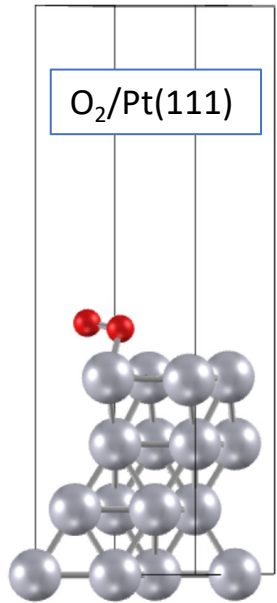
Note Sheet

	A	B	C	D	E	F	G	H	I	J
1	Structure	Total Energy			Total Energy	Adsorption Energy				
2	H2	-31.73444								
3	H2O	-599.14416		H2O/Pt(111)	-163274.22	0.0746199				
4	O2	-1129.8111		O2/Pt(111)	-163805.15	0.3390900				
5	OOH*	-1145.6533		OOH/Pt(111)	-163821.48	0.8269900				
6	O*	-561.8097		O/Pt(111)	-163240.61	3.7857800				
7	OH*	-580.16305		OH/Pt(111)	-163257.31	=-(E7-(B7+B8))				
8	Pt(111)	-162675.00								
9										
10										
11										

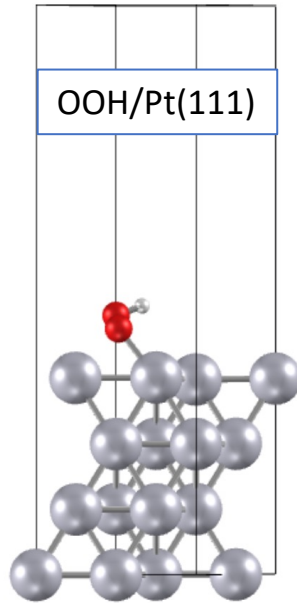
[Lab.3] Oxygen Reduction Reaction (ORR)

8. Calculate adsorption energies

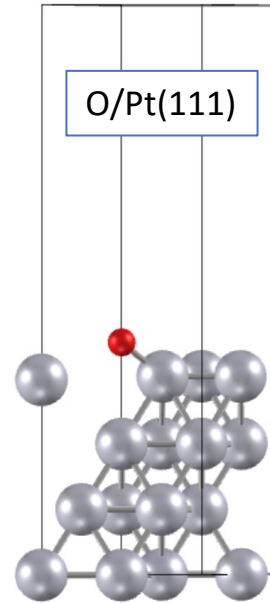
$$E_{\text{adsorption}} = -[E_{\text{adsorption structure}} - (E_{\text{molecule}} + E_{\text{surfaces}})]$$



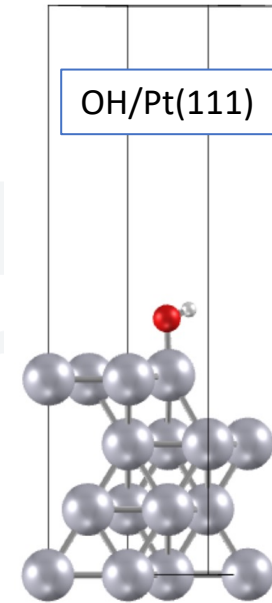
$E_{\text{ads}} = 0.34 \text{ eV}$



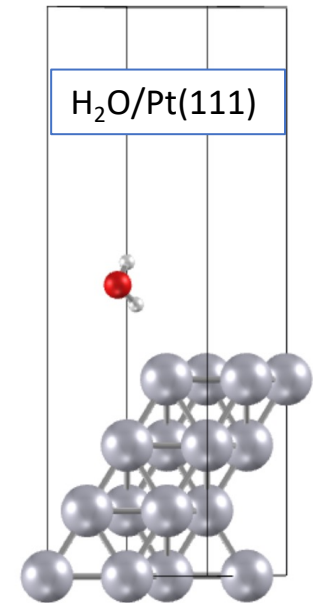
$E_{\text{ads}} = 0.83 \text{ eV}$



$E_{\text{ads}} = 3.79 \text{ eV}$



$E_{\text{ads}} = 2.20 \text{ eV}$



$E_{\text{ads}} = 0.00 \text{ eV}$

[Lab.3] Oxygen Reduction Reaction (ORR)

9. Calculate Free-energy diagram

- $\Delta E_{\text{OOH}^*} = E(\text{OOH}^*/\text{slab}) - E(\text{slab}) - \left\{ 2E(\text{H}_2\text{O}) - \frac{3}{2}E(\text{H}_2) \right\}$
- $\Delta E_{\text{O}^*} = E(\text{O}^*/\text{slab}) - E(\text{slab}) - \{E(\text{H}_2\text{O}) - E(\text{H}_2)\}$
- $\Delta E_{\text{OH}^*} = E(\text{OH}^*/\text{slab}) - E(\text{slab}) - \left\{ E(\text{H}_2\text{O}) - \frac{1}{2}E(\text{H}_2) \right\}$

Memo ▲

Note Sheet

	A ▼	B ▼	C ▼	D ▼	E ▼	F ▼	G ▼	H ▼	I ▼	J ▼
1	Structure	Total Energy			Total Energy	Adsorption Energy	Delta Energy			
2	H2	-31.73444								
3	H2O	-599.14416		H2O/Pt(111)	-163274.2	0.0746199	-0.0746199			
4	O2	-1129.8117		O2/Pt(111)	-163805.1	0.3390900	4.6692499			
5	OOH*	-1145.6533		OOH/Pt(111)	-163821.4	0.8269900	4.2062800			
6	O*	-561.80977		O/Pt(111)	-163240.6	3.7857800	1.8142299			
7	OH*	-580.16305		OH/Pt(111)	-163257.3	2.1999200	=E7-B8-(B3-0.5*B2)			
8	Pt(111)	-162675.00								
9										
10										
11										

[Lab.3] Oxygen Reduction Reaction (ORR)

9. Calculate Free-energy diagram

- $\Delta E_{\text{OOH}^*} = E(\text{OOH}^*/\text{slab}) - E(\text{slab}) - \left\{ 2E(\text{H}_2\text{O}) - \frac{3}{2}E(\text{H}_2) \right\}$
- $\Delta E_{\text{O}^*} = E(\text{O}^*/\text{slab}) - E(\text{slab}) - \{E(\text{H}_2\text{O}) - E(\text{H}_2)\}$
- $\Delta E_{\text{OH}^*} = E(\text{OH}^*/\text{slab}) - E(\text{slab}) - \left\{ E(\text{H}_2\text{O}) - \frac{1}{2}E(\text{H}_2) \right\}$

